Toric Code Demystified

From the Abelian Higgs Lagrangian to a Spin Model on Links

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Abstract

These notes derive the toric code, without shortcuts, from the abelian Higgs model. Part 1 carries the continuum theory through the Dirac procedure: two first-class constraints appear, and the choice of how to fix them separates the temporal presentation from the Coulomb one. Part 2 explains why only the temporal form can be discretized, performs the discretization with the canonical commutator carried across explicitly, reduces the gauge group to $\mathbb{Z}_2$, and arrives at the Fradkin–Shenker model. Its $t = J = 0$ corner is the toric code, with the vertex operator inherited from the matter's charging energy. Part 3 shows that no local order parameter distinguishes the regions of its phase diagram (Elitzur's theorem), describes what replaces one, and identifies the four-fold degeneracy of the toric code on a torus with the $\mathbb{Z}_2$ holonomies.

Companion note. For the Dirac–Bergmann formalism, gauge fixing and a planned comparison of quantization schemes, see *Quantization of Constrained Systems*.

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1 Constrained dynamics of the abelian Higgs model

1.1 Lagrangian and primary constraint

The abelian Higgs model — equivalently scalar electrodynamics, or relativistic Ginzburg–Landau — is

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \big(D_\mu\phi\big)^*\big(D^\mu\phi\big) - V\big(|\phi|^2\big), \qquad D_\mu = \partial_\mu - ieA_\mu. \tag{1}$$

In three-vector notation, with $\vec{E} = \dot{\vec{A}} - \nabla A_0$ and $\vec{B} = \nabla \times \vec{A}$,

$$\mathcal{L} = \tfrac{1}{2}\big(E^2 - B^2\big) + \big|D_0\phi\big|^2 - \big|\vec{D}\phi\big|^2 - V. \tag{2}$$

The covariant derivative acts in the time direction as well, $D_0\phi = \dot{\phi} - ieA_0\phi$, and it is worth following where that A_0 ends up: not in the energy, but as the coefficient of a constraint.

1.1.1 Momenta

$$\pi_i = \frac{\partial\mathcal{L}}{\partial\dot{A}_i} = E_i, \qquad \pi_\phi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \big(D_0\phi\big)^*, \qquad \boxed{\pi^0 = \frac{\partial\mathcal{L}}{\partial\dot{A}_0} = 0} \tag{3}$$

The last is not an equation to be solved but a *primary constraint* [Dir64; HT92],

$$\Phi_1 := \pi^0 \approx 0, \tag{4}$$

and it arises because $F_{\mu\nu}$ is antisymmetric, so $\dot{A}_0$ never appears in (1).

1.1.2 The Legendre transform of a singular Lagrangian

Remark 1.1 (why a multiplier appears)**.** The Legendre transform is performed as usual,

$$H = \pi^0\dot{A}_0 + \pi^i\dot{A}_i + \pi_\phi\dot{\phi} + \text{c.c.} - \mathcal{L}, \tag{5}$$

but the map $\dot{q} \mapsto p$ is not invertible: the relations for π^i, π_ϕ can be solved for $\dot{A}_i, \dot{\phi}$, whereas $\pi^0 = 0$ carries no information about $\dot{A}_0$. Since $\mathcal{L}$ contains no $\dot{A}_0$, all remaining terms are independent of it, so

$$H = \underbrace{\big[\pi^i\dot{A}_i + \pi_\phi\dot{\phi} + \text{c.c.} - \mathcal{L}\big]}_{H_c(q,p)} + \dot{A}_0\,\pi^0, \tag{6}$$

and $\dot{A}_0$ is simply not determined by the phase-space data. Renaming it $\lambda_1 := \dot{A}_0$ gives the total Hamiltonian. The term is not inserted by hand; it is the $\pi^0\dot{A}_0$ term of the Legendre transform, with the undetermined velocity kept as an unknown. Consistency is immediate: $\dot{A}_0 = \{A_0, H_T\} = \lambda_1$.

$$H_c = \int \mathrm{d}^3x\, \Big[\, \underbrace{|\pi_\phi|^2 + \big|\vec{D}\phi\big|^2 + V}_{\text{matter}} + \underbrace{\tfrac{1}{2}\big(E^2 + B^2\big)}_{\text{gauge field}} \,\Big] \;-\; \int \mathrm{d}^3x\; A_0\,\big(\nabla\cdot E - \rho\big), \tag{7}$$

$$H_T = H_c + \int \mathrm{d}^3x\; \lambda_1\,\pi^0, \qquad \rho = ie\big(\phi\,\pi_\phi - \phi^*\pi_\phi^*\big). \tag{8}$$

1.2 The second constraint, and why both are first class

Proposition 1.2. *Consistency of Φ_1 produces the Gauss constraint*

$$\Phi_2 := \nabla\cdot E - \rho \approx 0, \tag{9}$$

and the procedure terminates there.

Proof. $\dot{\Phi}_1 = \{\pi^0, H_T\} = -\,\partial H_T/\partial A_0 = \nabla\cdot E - \rho$. That $\dot{\Phi}_2 = 0$ identically is Proposition 1.3 below. □

Proposition 1.3. *Φ_1 and Φ_2 are first class, and $\{\Phi_2, H_T\} = 0$ identically.*

Proof. Smear the Gauss constraint, $G[\alpha] = \int \alpha\,\Phi_2$. Reading off its Poisson brackets (integrating $\int \alpha\,\partial_j E_j$ by parts),

$$\{A_i, G\} = -\partial_i\alpha, \quad \{E_i, G\} = 0, \quad \{\phi, G\} = -ie\,\alpha\phi, \quad \{\pi_\phi, G\} = +ie\,\alpha\pi_\phi, \tag{10}$$

so $G[\alpha]$ generates precisely a $U(1)$ gauge transformation. Under it

$$\delta\big(D_i\phi\big) = \partial_i(\delta\phi) - ie(\delta A_i)\phi - ieA_i\,\delta\phi = -ie\,\alpha\,D_i\phi, \tag{11}$$

which is covariance, so $|D_i\phi|^2$ is invariant; $|\pi_\phi|^2$ is invariant because ϕ and π_ϕ carry opposite phases; $V(|\phi|^2)$ because $|\phi|^2$ is inert; E because $\delta E = 0$; and B because $\delta B = \nabla\times(-\nabla\alpha) = 0$. Hence the first integral of (7) has vanishing bracket with G. For the second, A_0 commutes with everything in G, and $\{\Phi_2(x), \Phi_2(y)\} = 0$ since the only potentially nonzero piece, $\{\rho(x), \rho(y)\}$, is proportional to $-\phi(x)\pi_\phi(y)\delta + \phi(y)\pi_\phi(x)\delta$, which vanishes at $x = y$. Finally $\{\Phi_1, \Phi_2\} = 0$ because Φ_2 contains no A_0. □

Remark 1.4. Two first-class constraints remove $2\times 2 = 4$ from the eight phase-space dimensions of (A_μ, π^μ), leaving four: two polarizations. Each first-class constraint removes two dimensions, one for the constraint and one for the gauge orbit it generates.

1.3 Fixing one constraint: temporal gauge

Impose the single gauge condition

$$\chi_1 := A_0 \approx 0. \tag{12}$$

Since $\{\Phi_1, \chi_1\} = -1$, the pair (Φ_1, χ_1) is second class and (π^0, A_0) may be eliminated.

Proposition 1.5. *Eliminating (π^0, A_0) changes no bracket among the remaining variables.*

Proof. For F, G built from $(A_i, E_i, \phi, \pi_\phi)$ one has $\{F, \pi^0\} = \{F, A_0\} = 0$, so the Dirac correction $\{F, \Xi_a\}C^{ab}\{\Xi_b, G\}$ vanishes term by term. □

Setting $A_0 = 0$ in (7) deletes the last integral, leaving the familiar

$$\boxed{H = \int \mathrm{d}^3x\,\Big[\,|\pi_\phi|^2 + \big|\vec{D}\phi\big|^2 + V + \tfrac{1}{2}\big(E^2 + B^2\big)\,\Big]} \tag{13}$$

with Φ_2 still first class and therefore imposed on states,

$$\big(\nabla\cdot\hat{E} - \hat{\rho}\big)|\psi\rangle = 0. \tag{14}$$

The residual gauge freedom is a time-independent $\alpha(\vec{x})$, generated by Φ_2.

Remark 1.6. The phase-space variable A_0 has no tendency to move: $\dot{A}_0 = \lambda_1$, and $\lambda_1 = 0$ keeps $A_0 = 0$ for all time. Nothing in the dynamics pushes it. In particular Φ_2 cannot move it, since $\{A_0, \Phi_2\} = 0$.

This is the form one discretizes, because it is local and because (14) becomes a condition at each lattice site.

Remark 1.7 (temporal and Coulomb conditions together)**.** Temporal gauge sets $A_0 = 0$ and Coulomb gauge sets $\nabla\cdot A = 0$; using both at once is called radiation gauge. Under (8), $\dot{\vec{A}} = \vec{E} + \nabla A_0$, so on the Gauss constraint

$$\partial_t\big(\nabla\cdot A\big) - \nabla^2 A_0 = \rho. \tag{15}$$

Setting both terms to zero forces $\rho = 0$: with charges present the two conditions cannot hold together. The temporal form used below keeps $A_0 = 0$ and imposes Gauss's law on states, (14). The full analysis — the Coulomb drift, Dirac brackets, the extended Hamiltonian and the counting of multipliers — is in the companion note *Quantization of Constrained Systems*.

Remark 1.8 (what to discretize)**.** The temporal form (13) is the one to put on a lattice: it is local, and the surviving first-class constraint becomes a condition at each site. Writing $\phi_v = \mathrm{e}^{i\theta_v}$ with conjugate integer charge n_v, and A_ℓ on links with conjugate E_ℓ, Gauss's law becomes the discrete divergence

$$\sum_{\ell\ni v} E_\ell = n_v, \tag{16}$$

and for gauge group $\mathbb{Z}_2$, exponentiating by $\mathrm{e}^{i\pi(\cdot)}$ turns this into $\prod_{\ell\ni v}\sigma^x_\ell = \tau^x_v$, the familiar vertex condition of lattice $\mathbb{Z}_2$ gauge theory with matter. The Coulomb form (18) is unsuitable for that purpose, being nonlocal.

2 Lattice regularization and the $\mathbb{Z}_2$ reduction

Section 1 arrived at the temporal presentation of the theory; a Coulomb presentation is also available (Remark 1.7). We now explain why exactly one of them can be discretized, carry out the discretization, and reduce the gauge group to $\mathbb{Z}_2$. What emerges is the Fradkin–Shenker model [FS79], and at one corner of its parameter space, the toric code.

2.1 Why the temporal form

The temporal form of §1 and the Coulomb form — obtained by also imposing $\nabla\cdot A = 0$ and solving Gauss's law, so that the longitudinal electric field becomes the instantaneous Coulomb interaction — are

$$H_{\text{temporal}} = \int \mathrm{d}^3x\,\Big[|\pi_\phi|^2 + \big|\vec{D}\phi\big|^2 + V + \tfrac{1}{2}\big(E^2 + B^2\big)\Big], \qquad \big(\nabla\cdot\hat{E} - \hat{\rho}\big)|\psi\rangle = 0, \tag{17}$$

$$H_{\text{Coulomb}} = \int \mathrm{d}^3x\,\Big[\cdots + \tfrac{1}{2}\big(E_T^2 + B^2\big)\Big] + \frac{1}{8\pi}\iint \frac{\rho(x)\rho(y)}{|x-y|}, \qquad \text{no constraint.} \tag{18}$$

Three properties single out (17).

(i) **Locality.** The Coulomb term in (18) couples every pair of points. A lattice Hamiltonian must have finite range, so (18) cannot be discretized as a local model at all.

(ii) **The constraint stays local.** Because Φ_2 was left first class, it survives as a condition at each point, and under discretization it becomes one condition per vertex.

(iii) **Only space needs discretizing.** A_0 has been eliminated, so no time-direction structure is required. This is the Hamiltonian lattice framework, as opposed to the Wilson action formulation, which discretizes spacetime.

2.2 From Poisson brackets to link operators

2.2.1 Quantization

By Proposition 1.5 of §1, eliminating (π^0, A_0) leaves the remaining brackets untouched, so

$$\{A_i(\vec{x}), E_j(\vec{y})\} = \delta_{ij}\,\delta^3(\vec{x}-\vec{y}) \qquad \xrightarrow{\{,\}\to\frac{1}{i}[,]} \qquad \big[\hat{A}_i(\vec{x}), \hat{E}_j(\vec{y})\big] = i\,\delta_{ij}\,\delta^3(\vec{x}-\vec{y}). \tag{19}$$

2.2.2 Smearing: the delta function becomes a Kronecker delta

Introduce a cubic lattice of spacing a. The two fields are smeared over dual objects: $\vec{A}$ along links, $\vec{E}$ across the dual plaquettes that the links pierce,

$$\theta_\ell := g\int_\ell \vec{A}\cdot \mathrm{d}\vec{l} \;\approx\; g\,a\,A_i, \qquad E_\ell := \int_{S_\ell} \vec{E}\cdot \mathrm{d}\vec{S} \;\approx\; a^2 E_i, \tag{20}$$

where S_ℓ is the dual plaquette transverse to ℓ.

Proposition 2.1. *With the smearing* (20)*,*

$$\big[\theta_\ell,\, E_{\ell'}\big] = i\,g\,\delta_{\ell\ell'}. \tag{21}$$

Proof. Insert (19):

$$\big[\theta_\ell, E_{\ell'}\big] = i\,g\int_\ell \mathrm{d}x_i \int_{S_{\ell'}} \mathrm{d}S_j\;\delta_{ij}\,\delta^3(x-y). \tag{22}$$

The remaining integral counts the number of times the curve ℓ pierces the surface $S_{\ell'}$, with sign. A link pierces its own dual plaquette exactly once and every other dual plaquette zero times. □

Defining $L_\ell := E_\ell/g$ gives the canonical pair

$$\boxed{\big[\theta_\ell,\, L_{\ell'}\big] = i\,\delta_{\ell\ell'}} \tag{23}$$

Remark 2.2 (the geometry is the point)**.** Nothing here is a heuristic replacement of δ^3 by $\delta_{\ell\ell'}$. The Kronecker delta is the intersection number of a link with a dual plaquette, and it is what forces the gauge field onto links and the electric field onto the dual lattice.

2.2.3 Compactification and the integer spectrum

Definition 2.3. The gauge group is taken compact: $\theta_\ell \equiv \theta_\ell + 2\pi$, with holonomy $U_\ell := \mathrm{e}^{i\theta_\ell} \in U(1)$.

Because θ_ℓ is an angle, its conjugate has integer spectrum, $L_\ell \in \mathbb{Z}$, and

$$U_\ell^\dagger\, L_\ell\, U_\ell = L_\ell + 1, \tag{24}$$

which follows from (23): $\mathrm{e}^{-i\theta} L\, \mathrm{e}^{i\theta} = L - i[\theta, L] = L + 1$. The holonomy is the operator that lays down one unit of electric flux.

Remark 2.4. Compactness is a genuine choice, not a consequence. Taking $\theta_\ell \in \mathbb{R}$ gives a continuous electric spectrum and no confining phase. Compactness is what will make the reduction to $\mathbb{Z}_2$ available at all.

2.3 The Kogut–Susskind Hamiltonian

2.3.1 Electric term

With $E_i \approx gL_\ell/a^2$ and $\int \mathrm{d}^3x \to a^3 \sum$,

$$\int \tfrac{1}{2}E^2 \;\longrightarrow\; a^3 \sum \tfrac{1}{2}\Big(\frac{g\,L_\ell}{a^2}\Big)^2 \;=\; \frac{g^2}{2a}\sum_\ell L_\ell^2. \tag{25}$$

2.3.2 Magnetic term, and where B^2 actually comes from

The lattice curl is the holonomy around the smallest closed loop,

$$\Phi_p := \sum_{\ell\in\partial p} \theta_\ell = g \oint_{\partial p} \vec{A}\cdot \mathrm{d}\vec{l} \;\approx\; g\,a^2 B, \qquad W_p := \prod_{\ell\in\partial p} U_\ell = \mathrm{e}^{i\Phi_p}. \tag{26}$$

W_p is not Hermitian, so it cannot be a term in H by itself. Adding the oppositely oriented loop,

$$-\tfrac{1}{2}\big(W_p + W_p^\dagger\big) = -\cos\Phi_p = -1 + \tfrac{1}{2}\Phi_p^2 - \tfrac{1}{24}\Phi_p^4 + \cdots \tag{27}$$

Remark 2.5 (Hermiticity is what produces B^2)**.** The linear term $i\Phi_p \sim iga^2B$ in the expansion of $\mathrm{e}^{i\Phi_p}$ is imaginary and cancels against its conjugate. It is Hermiticity, not the expansion by itself, that leaves $\frac{1}{2}\Phi_p^2$ as the leading term. Reinstating the prefactor,

$$\int \tfrac{1}{2}B^2 \;\longrightarrow\; a^3 \sum \frac{\Phi_p^2}{2g^2a^4} \quad\Longrightarrow\quad H_B = \frac{1}{g^2 a}\sum_p \big(1-\cos\Phi_p\big). \tag{28}$$

Remark 2.6 (what is forced and what is not)**.** Gauge invariance forces θ_ℓ to appear only in closed loops (or terminating on matter, §2.3.3); Hermiticity forces the cos; the naive continuum limit forces the normalization. Which loop is used is *not* forced — larger loops have the same continuum limit, and improved actions exploit this. The plaquette is simply the smallest.

2.3.3 Matter, and the London limit

Write $\phi = |\phi|\mathrm{e}^{i\theta}$ and freeze the amplitude at the minimum of V.

Definition 2.7 (London limit)**.** Set $|\phi| = v$ with v fixed, retaining only the phase θ as a dynamical variable. Formally this is $\lambda \to \infty$ at fixed v in $V = \lambda(|\phi|^2 - v^2)^2$.

Remark 2.8 (when this is legitimate)**.** The amplitude mode has a gap $m_H \sim \sqrt{\lambda}\,v$ while the phase mode does not. For $E \ll m_H$ the amplitude cannot be excited and is a spectator. The limit is therefore accurate deep inside the ordered phase and fails (i) near the transition, where amplitude fluctuations are critical, and (ii) at vortex cores, where $|\phi| \to 0$ is precisely what allows the phase to wind. In a superconductor the limit is the standard London description: $|\Delta|$ is rigid and only its phase fluctuates.

The surviving pair is phase and charge,

$$\big[\theta_v,\, n_v\big] = i, \qquad n_v \in \mathbb{Z}, \tag{29}$$

with $\mathrm{e}^{i\theta_v}$ raising n_v by one — it is the operator that creates a boson. The two matter terms become

$$\int |\pi_\phi|^2 \;\longrightarrow\; \frac{U}{2}\sum_v n_v^2, \qquad \int \big|\vec{D}\phi\big|^2 \;\longrightarrow\; -t\sum_\ell \cos\big(\theta_v - \theta_{v'} + \theta_\ell\big), \tag{30}$$

with $t \sim v^2 a$, the link ℓ being oriented from v to v'.

Remark 2.9 (why θ_ℓ sits between the two phases)**.** ϕ_v and $\phi_{v'}$ live in different fibres; the gauge group acts independently at each site, so $\phi_v^*\phi_{v'}$ is not even well defined. Comparing them requires a parallel transporter, and the link holonomy is exactly that over one lattice spacing, $\phi^*(y)\mathrm{e}^{ig\int_x^y A}\phi(x)$. Equation (30) is the covariant derivative, not a repair.

2.3.4 Gauss law

The constraint of (17) is smeared over the closed surface bounding a vertex, giving a discrete divergence:

$$\nabla\cdot E=\rho \qquad \longrightarrow \qquad \boxed{\sum_{\ell\ni v} L_\ell = n_v} \tag{31}$$

Here each link is counted with $+$ if it is oriented out of v and $-$ if into it, and likewise Φ_p is the oriented sum around ∂p; for $\mathbb{Z}_2$ the signs are immaterial.

Collecting (25), (28), (30), the Kogut–Susskind Hamiltonian [KS75]:

$$H=\frac{g^2}{2a}\sum_\ell L_\ell^2-\frac{1}{g^2 a}\sum_p\cos\Phi_p+\frac{U}{2}\sum_v n_v^2-t\sum_\ell\cos\left(\theta_v-\theta_{v'}+\theta_\ell\right). \tag{32}$$

Remark 2.10. The generators of (31) mutually commute, commute with (32), and sum to zero on a torus; all three were checked numerically on a 2×2 lattice with the electric field truncated to $L\in\{-1,0,1\}$.

2.4 Reduction to $\mathbb{Z}_2$

2.4.1 What the reduction is

Replacing $U(1)$ by $\mathbb{Z}_2$ restricts

$$\theta_\ell\in[0,2\pi) \;\longrightarrow\; \{0,\pi\}, \qquad L_\ell\in\mathbb{Z} \;\longrightarrow\; \{0,1\}, \tag{33}$$

and likewise for the matter. Since $\mathbb{Z}_2$ is its own Pontryagin dual, each link and each site carries a two-dimensional Hilbert space with two conjugate bases,

$$\theta\in\{0,\pi\} \qquad \xleftrightarrow{\mathbb{Z}_2\ \text{Fourier}} \qquad L\in\{0,1\}. \tag{34}$$

Remark 2.11 (this is not a limit of the continuum theory)**.** $U(1)\to\mathbb{Z}_2$ changes the gauge group; it is not an approximation to scalar electrodynamics. The physical route to it is condensation: a charge-2 condensate breaks $U(1)\to\mathbb{Z}_2$ and the low-energy theory is a $\mathbb{Z}_2$ gauge theory. That is the superconductor, with e the Bogoliubov quasiparticle and m the vortex.

2.4.2 Pauli operators are forced, not substituted

Proposition 2.12. *On the two-dimensional space, the operators*

$$\sigma_\ell^z:=\mathrm{e}^{i\theta_\ell}, \qquad \sigma_\ell^x:=\mathrm{e}^{i\pi L_\ell} \tag{35}$$

satisfy $(\sigma^z)^2=(\sigma^x)^2=\mathbf{1}$ *and* $\{\sigma^z,\sigma^x\}=0$*, hence generate the Pauli algebra; and in the* θ *basis they are the standard matrices.*

Proof. Squares are immediate. For the anticommutator, (23) gives $\mathrm{e}^{i\pi L}\mathrm{e}^{i\theta}\mathrm{e}^{-i\pi L}=\mathrm{e}^{i(\theta+\pi)}=-\mathrm{e}^{i\theta}$, since $\mathrm{e}^{i\pi L}$ translates θ by π. In the θ basis $\mathrm{e}^{i\theta}=\mathrm{diag}(1,-1)$ and $\mathrm{e}^{i\pi L}$ is the exchange of the two states. □

Remark 2.13. The values ±1 are eigenvalues, not numbers replacing operators. What is being said is that a conjugate pair $[\theta,L]=i$, restricted to a two-dimensional space, can only be the Pauli algebra. The same statement on sites gives $\tau_v^z=\mathrm{e}^{i\theta_v}$, $\tau_v^x=\mathrm{e}^{i\pi n_v}$.

2.4.3 Term by term

Electric. For $L \in \{0,1\}$ one has $L^2 = L$, and $\sigma^x = \mathrm{e}^{i\pi L} = 1 - 2L$ gives $L = (1-\sigma^x)/2$, so

$$\frac{g^2}{2a}\sum_\ell L_\ell^2 = \text{const} - J\sum_\ell \sigma_\ell^x, \qquad J = \frac{g^2}{4a}. \tag{36}$$

Magnetic. $W_p = \prod_{\ell\in\partial p}\sigma_\ell^z =: B_p$ is already Hermitian, so the Hermitization of Remark 2.5 is trivial and

$$-\cos\Phi_p = -\tfrac{1}{2}\big(B_p + B_p\big) = -B_p. \tag{37}$$

Note that $\Phi_p \in \{0,\pi\}$ admits no small-Φ expansion: $-B_p$ is the energy of one unit of $\mathbb{Z}_2$ flux (a vison), not a discretization of $\frac{1}{2}B^2$. The $\mathbb{Z}_2$ deconfined phase is gapped and topological and has no Maxwell continuum limit.

Charging. Identically, $n^2 = n = (1-\tau^x)/2$ and $\frac{U}{2}\sum n_v^2 = \text{const} - \mu\sum_v \tau_v^x$ with $\mu = U/4$.

Hopping. Since all phases are ±1, $\mathrm{e}^{-ix} = \mathrm{e}^{+ix}$, so

$$\mathrm{e}^{i(\theta_v - \theta_{v'} + \theta_\ell)} = \mathrm{e}^{i\theta_v}\mathrm{e}^{i\theta_{v'}}\mathrm{e}^{i\theta_\ell} = \tau_v^z\,\sigma_\ell^z\,\tau_{v'}^z, \tag{38}$$

which is real, so again the $\frac{1}{2}$ of the Hermitization cancels and $\cos(\cdots) = \tau_v^z\sigma_\ell^z\tau_{v'}^z$.

Remark 2.14 (what the hopping operator does)**.** In the conjugate (charge and flux) basis, $\tau_v^z\sigma_\ell^z\tau_{v'}^z$ becomes $\tau_v^x\sigma_\ell^x\tau_{v'}^x$: it flips the charge at v, the flux on ℓ, and the charge at v' *together*. That is exactly what Gauss's law requires, and it is why the bare matter field τ_v^z and the bare link variable σ_ℓ^z are separately not gauge invariant while their dressed product is.

2.4.4 Gauss law, exponentiated

Proposition 2.15. *The finite gauge transformation at a vertex is the exponentiated Gauss generator,*

$$\mathrm{e}^{i\pi\hat{G}_v} = \prod_{\ell\ni v}\mathrm{e}^{i\pi L_\ell}\cdot\mathrm{e}^{-i\pi n_v} = \tau_v^x\prod_{\ell\ni v}\sigma_\ell^x \;=:\; G_v, \tag{39}$$

and it acts by

$$G_v\,\tau_v^z\,G_v = -\tau_v^z, \qquad G_v\,\sigma_\ell^z\,G_v = -\sigma_\ell^z \;\;(\ell\ni v), \qquad G_v\,\sigma_{\ell'}^z\,G_v = +\sigma_{\ell'}^z \;\;(\ell'\not\ni v). \tag{40}$$

Proof. Classically $\delta_\alpha F = \{F, G[\alpha]\}$; quantizing, $\delta_\alpha\hat{F} = \frac{1}{i}[\hat{F},\hat{G}]$, so a finite transformation is conjugation by $\mathrm{e}^{i\hat{G}[\alpha]}$. Taking $\alpha_v = \pi$ at one vertex and using $\mathrm{e}^{-i\pi n} = \mathrm{e}^{+i\pi n}$ for integer n gives the stated product. Equation (40) follows by counting anticommutations: τ_v^z meets τ_v^x once, σ_ℓ^z with $\ell\ni v$ meets σ_ℓ^x once, while a link not touching v meets nothing. (Verified numerically by matrix exponentiation.) □

Remark 2.16 (only the finite form exists)**.** For $U(1)$ the constraint may be written infinitesimally, $\hat{G}[\alpha]|\psi\rangle = 0$, or finitely, $\mathrm{e}^{i\hat{G}}|\psi\rangle = |\psi\rangle$. For $\mathbb{Z}_2$ the group is discrete and only the finite form is available:

$$G_v|\psi\rangle = |\psi\rangle. \tag{41}$$

This is why lattice $\mathbb{Z}_2$ gauge theory is always presented with a vertex operator rather than a divergence.

Interpreting (40) in terms of α: from $\theta_\ell \to \theta_\ell - g(\alpha_{v'} - \alpha_v)$ and $\theta_v \to \theta_v - g\alpha_v$, writing $\epsilon_v = \mathrm{e}^{-ig\alpha_v} \in \{\pm1\}$,

$$\tau_v^z \to \epsilon_v\tau_v^z, \qquad \sigma_\ell^z \to \epsilon_v\epsilon_{v'}\sigma_\ell^z, \tag{42}$$

and choosing $\epsilon = -1$ at a single vertex reproduces (40) exactly.

2.5 The Fradkin–Shenker model

$$\boxed{H = -\mu \sum_v \tau_v^x \;-\; \sum_p B_p \;-\; t \sum_\ell \tau_v^z \sigma_\ell^z \tau_{v'}^z \;-\; J \sum_\ell \sigma_\ell^x} \qquad \mu = \frac{U}{4}, \quad J = \frac{g^2}{4a}, \tag{43}$$

with physical states obeying (41).

continuum	$U(1)$ lattice	$\mathbb{Z}_2$	meaning
$\frac{1}{2}E^2$	$\frac{g^2}{2a}L_\ell^2$	$-J\sigma_\ell^x$	cost of electric flux
$\frac{1}{2}B^2$	$\frac{1}{g^2 a}(1-\cos\Phi_p)$	$-B_p$	cost of a vison
$\lvert\pi_\phi\rvert^2$	$\frac{U}{2}n_v^2$	$-\mu\tau_v^x$	cost of a charge
$\lvert\vec{D}\phi\rvert^2$	$-t\cos(\theta_v-\theta_{v'}+\theta_\ell)$	$-t\,\tau^z\sigma^z\tau^z$	covariant hopping
$\nabla\cdot E=\rho$	$\sum_{\ell\ni v} L_\ell = n_v$	$G_v = 1$	Gauss law

2.6 Unitary gauge and the toric code

The matter in (43) is not an independent degree of freedom. Making that precise requires two separate statements, which are easily conflated.

2.6.1 Counting

Proposition 2.17. $\dim \mathcal{H}_{\text{phys}} = 2^{N_\ell}$, *where* $\mathcal{H}_{\text{phys}}$ *is the* $G_v = 1$ *subspace.*

Proof. The full space has dimension $2^{N_v+N_\ell}$ and the N_v conditions (41) are independent, since $\prod_v G_v = \prod_v \tau_v^x \neq \mathbf{1}$. On a 2×2 torus this gives $2^{12}/2^4 = 2^8 = 256$, confirmed by explicit projection. □

2.6.2 Statement A: the gauge slice

The gauge group $\mathbb{Z}_2^{N_v}$ acts on configurations $(\{\tau_v^z\},\{\sigma_\ell^z\})$ by (40). The action is *free*, since a nontrivial element flips at least one τ_v^z. Hence every orbit has 2^{N_v} points, there are 2^{N_ℓ} orbits, and each orbit contains *exactly one* configuration with all $\tau_v^z = +1$. This is a complete and unique gauge fixing of the configuration space.

2.6.3 Statement B: it is not a condition on physical states

Proposition 2.18. $P\,\tau_v^z\,P = 0$, *where* P *projects onto* $\mathcal{H}_{\text{phys}}$.

Proof. τ_v^z anticommutes with G_v by (40), so $P\tau_v^z P = P\tau_v^z G_v P = -PG_v\tau_v^z P = -P\tau_v^z P$. (Verified numerically.) □

Physical states are uniform superpositions over gauge orbits; no physical state has $\tau_v^z = +1$. What Statement A provides is therefore not a constraint but an *isomorphism*

$$W:\ \mathcal{H}_{\text{link}} \longrightarrow \mathcal{H}_{\text{phys}}, \qquad |\text{link}\rangle \longmapsto P\Big(|\tau^z{=}{+}1\rangle^{\otimes N_v} \otimes |\text{link}\rangle\Big), \tag{44}$$

which is well defined and unitary after normalization by Proposition 2.17. Unitary gauge is a choice of coordinates on $\mathcal{H}_{\text{phys}}$, not a reduction of it.

2.6.4 Transporting the Hamiltonian

Under W, the hopping term becomes $-t\sum_\ell \sigma^z_\ell$, since $\tau^z = +1$ on the slice. The charging term needs more care: τ^x_v flips τ^z_v and therefore leaves the slice. It must be dressed by a gauge transformation to return,

$$\tau^x_v\, G_v = \tau^x_v \cdot \tau^x_v \prod_{\ell \ni v} \sigma^x_\ell = \prod_{\ell \ni v} \sigma^x_\ell =: A_v, \tag{45}$$

and $G_v = \mathbf{1}$ on $\mathcal{H}_{\text{phys}}$, so τ^x_v acts there as A_v. Hence

$$\boxed{H = -\mu \sum_v A_v - \sum_p B_p - t\sum_\ell \sigma^z_\ell - J\sum_\ell \sigma^x_\ell} \tag{46}$$

on N_ℓ qubits with no constraint.

Remark 2.19 (numerical check)**.** For a 2×2 torus with $(\mu, t, J) = (0.9, 0.6, 0.4)$, the spectrum of (43) restricted to the 256-dimensional $\mathcal{H}_{\text{phys}}$ agrees with the full spectrum of (46) to 10^{-8}, level by level.

Remark 2.20 (this is not "integrating out")**.** Integrating out a degree of freedom generically produces new effective interactions. Here nothing of the sort occurs: the matter was never independent, and (46) is exact.

2.6.5 The toric code

Setting $t = J = 0$ in (46),

$$H = -\mu \sum_v A_v - \sum_p B_p, \tag{47}$$

the toric code [Kit03]. Two observations.

First, the vertex term *came from the matter*: it is the charging energy $-\mu\tau^x_v$, transported through unitary gauge. In the toric code as usually presented, A_v is postulated; here it has the meaning of the gap to a single e charge, $A_v = -1$ signalling a charge at v at cost 2μ.

Second, this makes precise a distinction that matters for what follows.

	Gauss law	states with $A_v = -1$
with matter (here)	$A_v = \tau^x_v$, exact	physical, a charge is present
pure gauge, energetic	$A_v = 1$ enforced by $-\mu\sum A_v$	excited, emergent Gauss law

Remark 2.21. The two columns give different physics — in particular the robustness of entanglement-spectrum degeneracies differs — so any calculation on (43) should state which is meant. The derivation above produces the first.

3 Order parameters for a gauge theory

Section 2 produced the Fradkin–Shenker model and identified the toric code as one corner of it. The question left open is what distinguishes the corners. This section shows that the usual answer — a local order parameter — is not merely inconvenient here but provably absent, and identifies what replaces it. Space is two-dimensional (a square lattice) from here on: the construction of §2 goes through unchanged except for the powers of a in the couplings, while the ground-state degeneracy and the self-duality quoted below are two-dimensional statements.

Throughout, statements are tagged [rigorous], [standard, proof omitted] or [heuristic]. Facts about convexity, clustering and superselection are used freely and are proved in the companion notes *Effective Potentials, Convexity, and the Definition of a Phase*, cited as [B].

3.1 The problem

The model (43) has three qualitatively different regions.

corner	phase	condensate	ground-state degeneracy on T^2
$t = J = 0$	deconfined (toric code)	none	4
t large	Higgs	e	1
J large	confined	m	1

The labels "Higgs" and "confined" name regimes, not separate phases: with matter of unit charge the two are connected without a phase transition [FS79] (Interpretation 3.29).

The Landau procedure would be to find a local field whose expectation value vanishes in one region and not in another. Theorem 3.8 below says no such field exists.

Remark 3.1 (self-duality)**.** [standard, proof omitted] On a self-dual lattice the model is invariant under exchanging vertices with plaquettes, $e \leftrightarrow m$ and $t \leftrightarrow J$. The Higgs and confined regions therefore sit symmetrically, which already suggests that no asymmetric criterion — such as "the matter field has condensed" — can be fundamental.

3.2 Haar averaging

The algebraic content of Elitzur's theorem is the following elementary fact.

Theorem 3.2 (Haar average is a projector)**.** *[rigorous] Let G be a compact group with normalized Haar measure $\mathrm{d}g$, and let ρ be a unitary representation on a finite-dimensional space W. Then*

$$\int_G \mathrm{d}g \; \rho(g) \;=\; P_{\mathrm{inv}},$$

the orthogonal projector onto the subspace $W^G = \{w : \rho(g)w = w \;\forall g\}$ of invariant vectors.

Proof. Set $M := \int \mathrm{d}g\, \rho(g)$. By left invariance of Haar measure, for any $h \in G$,

$$\rho(h)M = \int \mathrm{d}g \; \rho(hg) = \int \mathrm{d}g' \; \rho(g') = M,$$

so every vector in the image of M is invariant. Conversely if $w \in W^G$ then $Mw = \int \mathrm{d}g\, w = w$, using $\int \mathrm{d}g = 1$. Hence M acts as the identity on W^G and maps everything into W^G, so $M^2 = M$ and $\operatorname{im} M = W^G$. Self-adjointness follows from unitarity of ρ together with invariance of Haar measure under $g \mapsto g^{-1}$. □

Corollary 3.3. *[rigorous] If ρ contains no trivial subrepresentation then $\int \mathrm{d}g\, \rho(g) = 0$.*

Example 3.4. [rigorous] For $G = U(1)$ acting on a charge-q field, $\rho(\alpha) = \mathrm{e}^{iq\alpha}$ and $\int_0^{2\pi} \frac{\mathrm{d}\alpha}{2\pi} \mathrm{e}^{iq\alpha} = \delta_{q,0}$. For $G = \mathbb{Z}_2$ acting on an odd-charge field, $\frac{1}{2}(1 - 1) = 0$. For $G = \mathbb{Z}_N$ and charge $q \not\equiv 0$, $\frac{1}{N}\sum_k \mathrm{e}^{2\pi i k q/N} = 0$. A gauge-invariant operator carries the trivial representation and is untouched.

Caution 3.5 (compactness is essential)**.** The proof uses $\int \mathrm{d}g = 1$, which requires a normalizable invariant measure. For a non-compact gauge group no such measure exists and the conclusion fails. This is the same hypothesis that made the compactification of §2 a genuine choice rather than a formality.

3.3 Elitzur's theorem

Two statements travel under this name and they are not equally deep.

Proposition 3.6 (the kinematical version)**.** *[rigorous] Let $|\psi\rangle$ satisfy the Gauss law $G_v|\psi\rangle = |\psi\rangle$ and let $G_v\,\tau^z_v\,G_v^\dagger = -\tau^z_v$ as in Proposition 2.15. Then $\langle\psi|\tau^z_v|\psi\rangle = 0$.*

Proof. $\langle\psi|\tau^z_v|\psi\rangle = \langle\psi|G_v^\dagger\tau^z_v G_v|\psi\rangle = -\langle\psi|\tau^z_v|\psi\rangle$. □

Caution 3.7. Proposition 3.6 is close to a tautology: physical states were *defined* to be gauge invariant. It says nothing about whether a symmetry-breaking source could produce a condensate, which is the question Landau theory actually asks. That is the content of Theorem 3.8.

The nontrivial statement concerns the order of limits used to define spontaneous symmetry breaking in [B]: add an explicit breaking source h, take $V \to \infty$ first, then $h \to 0$.

Theorem 3.8 (Elitzur [Eli75], $\mathbb{Z}_2$ lattice gauge–Higgs)**.** *[rigorous] Consider the classical partition function on any finite lattice,*

$$Z_h = \sum_{\{\phi,\sigma\}} \mathrm{e}^{-S[\phi,\sigma]}\ \mathrm{e}^{h\sum_{v'}\phi_{v'}}, \qquad \phi_v, \sigma_\ell \in \{\pm 1\},$$

with S invariant under the local transformations of Proposition 2.15. Then for every site v,

$$\big\langle \phi_v \big\rangle_h \;=\; \tanh h,$$

independently of the couplings in S, of the lattice, of its dimension and of its volume. Consequently

$$\lim_{h\to 0}\ \lim_{V\to\infty}\ \big\langle \phi_v \big\rangle_h \;=\; 0.$$

Proof. Insert, for each configuration, an average over the gauge transformation $g_v \in \{\pm 1\}$ at the single site v:

$$\phi_v \mapsto g_v\phi_v, \qquad \sigma_\ell \mapsto g_v\sigma_\ell \quad (\ell \ni v).$$

This is a bijection of the summation variables, so the sum is unchanged; S is invariant by hypothesis; and among the source terms only the one at v involves ϕ_v. Hence

$$\big\langle \phi_v \big\rangle_h = \frac{1}{Z_h}\sum_{\{\phi,\sigma\}} \mathrm{e}^{-S}\ \mathrm{e}^{h\sum_{v'\neq v}\phi_{v'}} \cdot \underbrace{\tfrac{1}{2}\sum_{g_v=\pm 1} g_v\phi_v\,\mathrm{e}^{hg_v\phi_v}}_{(*)},$$

and correspondingly for Z_h with $(*)$ replaced by $\frac{1}{2}\sum_{g_v}\mathrm{e}^{hg_v\phi_v}$. Since $\phi_v = \pm 1$,

$$(*) = \tfrac{1}{2}\big(\phi_v\mathrm{e}^{h\phi_v} - \phi_v\mathrm{e}^{-h\phi_v}\big) = \phi_v\sinh(h\phi_v) = \sinh h, \qquad \tfrac{1}{2}\sum_{g_v}\mathrm{e}^{hg_v\phi_v} = \cosh h,$$

both independent of the configuration. They therefore factor out of numerator and denominator, and the remaining sums cancel:

$$\big\langle \phi_v \big\rangle_h = \frac{\sinh h}{\cosh h} = \tanh h.$$ □

Remark 3.9. Direct enumeration of a 2×2 torus at three different (β,κ,h) reproduces $\tanh h$ to eight decimals.

Interpretation 3.10 (why the local case is different)**.** The mechanism is a counting of costs, exactly parallel to [B, §4]. Flipping the source term at a single site changes the action by $O(h)$ regardless of volume; flipping all of them, as a global symmetry would, changes it by $O(hV)$. In the local case the site average factors out of the partition function and cancels; in the global case it does not, and the competition with V is what permits symmetry breaking.

	what is flipped	change in action	$\lim_{h\to 0}\lim_{V\to\infty}$
global symmetry	all sites	$O(hV)$	may be nonzero
gauge symmetry	one site	$O(h)$	0

Caution 3.11 (not Mermin–Wagner)**.** Theorem 3.8 holds in every dimension and at every temperature and coupling, because its proof is an exact cancellation and not an estimate of fluctuations. The Mermin–Wagner theorem is of a different type: it concerns continuous symmetries, relies on infrared divergences of Goldstone modes, and is dimension dependent. Compare [B, §4.5], where the analogous counting for a global discrete symmetry gives a barrier σL^{d-1} and hence a genuine dimension dependence.

3.4 Gauge fixing does not create an order parameter

Gauge fixing removes the redundancy that made Theorem 3.8 work, and after it $\langle\phi\rangle \neq 0$ is possible. The values obtained are, however, not physical.

Example 3.12 (two sites and one link)**.** [rigorous] Take matter τ_1, τ_2 and a link σ, with $G_1 = \tau_1^x\sigma^x$ and $G_2 = \tau_2^x\sigma^x$. In the basis $|s_1, m, s_2\rangle$ the gauge orbit of $|{+}{+}{+}\rangle$ is

$$|{+}{+}{+}\rangle, \quad |{-}{-}{+}\rangle, \quad |{+}{-}{-}\rangle, \quad |{-}{+}{-}\rangle,$$

all with $s_1 m s_2 = +1$, this being the gauge-invariant label; the other value of $s_1 m s_2$ gives the second orbit, so $\dim \mathcal{H}_{\text{phys}} = 2$. The physical state is the uniform superposition over an orbit, and

$$\langle\psi|\tau_1^z|\psi\rangle = \tfrac{1}{4}\big[(+1) + (-1) + (+1) + (-1)\big] = 0,$$

as Proposition 3.6 requires. Fixing the gauge selects one representative:

	$\langle\tau_2^z\rangle$	$\langle\sigma^z\rangle$	$\langle\tau_1^z\sigma^z\tau_2^z\rangle$
physical state (no gauge fixing)	0	0	+1
gauge $s_1 = s_2 = +$	+1	+1	+1
gauge $s_1 = +$, $s_2 = -$	−1	−1	+1

Interpretation 3.13. Only the last column is the same in all three rows. The quantity $\langle\tau^z\rangle$ takes the value $+1$ or -1 for the *same* physical state depending on a convention, so it is not an observable and cannot label a phase. Gauge fixing removes redundancy; it does not manufacture physical content.

Remark 3.14 (the Higgs mechanism is not symmetry breaking)**.** [standard, proof omitted] The statement "the gauge symmetry is broken" is, read literally, false. What is physical and gauge invariant is that the gauge boson acquires a mass, visible as the pole of a gauge-invariant correlator. The textbook formula $\langle\phi\rangle \neq 0$ is a statement in a fixed gauge, of the kind tabulated above.

3.5 What replaces the local order parameter

Since a charged local field has vanishing expectation value, a charge must be created by an extended operator with the charge at its ends.

Definition 3.15 (open string)**.** For a path γ from x to y,

$$\mathcal{O}_e(x,y) := \tau^z_x \Big(\prod_{\ell \in \gamma} \sigma^z_\ell \Big) \tau^z_y .$$

Proposition 3.16. *[rigorous] $\mathcal{O}_e(x,y)$ commutes with every G_v, whereas τ^z_x alone and $\prod_{\ell\in\gamma}\sigma^z_\ell$ alone do not.*

Proof. By Proposition 2.15, G_v anticommutes with τ^z_v and with σ^z_ℓ for each of the links touching v. An interior vertex of γ is touched by exactly two links of γ, giving two sign changes; an endpoint is touched by one link of γ and carries one τ^z, again two; a vertex off γ is touched by none. In every case the number of anticommutations is even. Removing either factor at an endpoint leaves an odd count there. □

Definition 3.17 (condensation)**.** The charge e is said to condense if the open string retains long-range order,

$$\lim_{|x-y|\to\infty} \big\langle \mathcal{O}_e(x,y) \big\rangle \;\neq\; 0,$$

equivalently if the string tension vanishes, i.e. the decay is perimeter-like rather than exponential in $|x-y|$.

Interpretation 3.18. This is the gauge-invariant transcription of $\langle\phi(x)\phi^*(y)\rangle \to |\langle\phi\rangle|^2$. Theorem 3.8 forbids splitting the two ends apart; what survives is the statement about the pair, connected by the string that makes it gauge invariant. In unitary gauge, where the matter has been eliminated (§2), the string reduces to $\prod_\gamma \sigma^z_\ell$ and the criterion becomes long-range order of $\prod_{\ell\in\gamma}\sigma^z_\ell$ as $|x-y|\to\infty$. A single $\langle\sigma^z_\ell\rangle$ does not discriminate: the term $-t\sum_\ell \sigma^z_\ell$ makes it nonzero for every $t \neq 0$.

Caution 3.19. Definition 3.17, the statement that the excitation gap closes, and the algebraic criterion of Definition 3.20 below are three characterizations that agree in the cases of interest but are not equivalent in general. A closing gap is necessary, not sufficient: condensation requires in addition that the vacuum reorganize into a superposition of arbitrarily long strings.

3.6 The algebraic criterion

Which sets of anyons *can* condense is fixed by consistency alone.

Definition 3.20 (Lagrangian algebra, abelian case)**.** [standard, proof omitted] A set $\mathcal{A}$ of anyons is condensable if

(i) every $a \in \mathcal{A}$ is a boson, $\theta_a = 1$;

(ii) all mutual braidings are trivial, $M_{ab} = 1$ for $a, b \in \mathcal{A}$;

(iii) $\mathcal{A}$ is closed under fusion.

It is Lagrangian if in addition

$$\sum_{a\in\mathcal{A}} d_a \;=\; \mathcal{D}, \qquad \mathcal{D} := \Big(\sum_{\text{all } b} d_b^2 \Big)^{1/2},$$

in which case the condensed theory is trivial.

Interpretation 3.21. Condensing means declaring the members of $\mathcal{A}$ to be identical to the vacuum. Condition (i) is required because the vacuum is bosonic — this is where Fermi statistics enters, and it is the reason $\{1, \psi\}$ is excluded below. Condition (ii) is required for the identification to be consistent under braiding, and (iii) for it to be consistent under fusion.

Proposition 3.22 (the toric code has exactly two Lagrangian algebras)**.** *[rigorous] For $\{1, e, m, \psi\}$ with $\theta_e = \theta_m = 1$, $\theta_\psi = -1$, $M_{em} = -1$ and $\mathcal{D} = 2$, the condensable Lagrangian algebras are*

$$\{1, e\} \qquad \text{and} \qquad \{1, m\}.$$

Proof. ψ is a fermion, so any $\mathcal{A}$ containing it violates (i). The set $\{1, e, m\}$ is not closed under fusion since $e \times m = \psi$, and $\{1, e, m, \psi\}$ fails (i); moreover $M_{em} = -1$ violates (ii). The remaining candidates containing a nontrivial element are $\{1, e\}$ and $\{1, m\}$, both of which satisfy (i)–(iii). All four anyons are abelian, $d_a = 1$, so $\mathcal{D} = \sqrt{4} = 2$, and each candidate has $\sum_{a \in \mathcal{A}} d_a = 1 + 1 = 2 = \mathcal{D}$: both are Lagrangian. □

Interpretation 3.23. The two algebras are the two ways of leaving the deconfined phase: condensing e (the Higgs regime) or m (the confined regime). That there are exactly two, and that both leave a trivial theory behind, is thus fixed before any microscopic calculation: the phase diagram of §2 could not have had a fourth region with residual topological order. Since both leave the same trivial theory and the model has no global symmetry, nothing requires the two regimes to be distinct phases, and they are not (Interpretation 3.29). Condensing e confines m, because dragging an m around the e-condensate accumulates the braiding phase $M_{em} = -1$ and hence a string tension.

3.7 The deconfined corner: degeneracy is holonomy

Proposition 3.24 (toric code degeneracy)**.** *[rigorous] On a torus, the toric code $H = -\mu \sum_v A_v - \sum_p B_p$ has a four-fold ground space, and the four states are distinguished by the eigenvalues of the two non-contractible Wilson loops*

$$W(\gamma) = \prod_{\ell \in \gamma} \sigma^z_\ell .$$

Proof. [standard, proof omitted] for the counting: on an $L_x \times L_y$ torus there are $N_\ell = 2L_xL_y$ links, L_xL_y vertex operators and L_xL_y plaquette operators, subject to the two relations $\prod_v A_v = \mathbf{1}$ and $\prod_p B_p = \mathbf{1}$; hence $2L_xL_y - 2$ independent stabilizers and $2^{N_\ell}/2^{N_\ell - 2} = 4$. [rigorous] for the labels: $W(\gamma)$ for non-contractible γ commutes with every A_v (it meets each vertex in an even number of links) and with every B_p (both are built from σ^z), so it acts within the ground space; the two loops commute there; and their joint eigenvalues realize all four sign pairs. Explicit diagonalization on a 2×2 torus confirms the degeneracy 4, the gap 4, and the four sign pairs. □

Interpretation 3.25 (the global data gauge fixing leaves)**.** $W(\gamma) = \exp\left(i \oint_\gamma A\right)$ is the $\mathbb{Z}_2$ holonomy. Gauge fixing removes local redundancy but cannot remove global data: the total charge and the holonomies around non-contractible cycles. On a torus there are two independent cycles and the gauge group is $\mathbb{Z}_2$, giving $2 \times 2 = 4$ sectors. The "topological" degeneracy of the toric code is therefore not a new phenomenon but the count of exactly those leftover variables.

Remark 3.26 (anyons are charges and fluxes)**.** [standard, proof omitted] In the same dictionary, $A_v = -1$ is an electric charge and $B_p = -1$ a unit of magnetic flux; their mutual statistics $M_{em} = -1$ is the Aharonov–Bohm phase $\mathrm{e}^{i \oint A} = \mathrm{e}^{i\pi}$ of a $\mathbb{Z}_2$ charge transported around a $\mathbb{Z}_2$ flux. Nothing about the braiding is more exotic than that.

Remark 3.27 (phase versus sector)**.** The four ground states are locally indistinguishable — no local operator connects or separates them — yet they are a bulk property depending on the topology of space. In the terminology of [B, §6.3] they are distinct superselection sectors that nonetheless constitute a single phase in the thermodynamic sense, and it is precisely this case that motivates enlarging the Landau framework.

3.8 Why the Wilson loop is not enough

The natural gauge-invariant diagnostic is the behaviour of loop operators; on the axes $J = 0$ (for the Higgs row) and $t = 0$ (for the confined row) it is:

	$\langle W_e \rangle = \langle \prod_\gamma \sigma^z \rangle$	$\langle W_m \rangle = \langle \prod_{\tilde{\gamma}} \sigma^x \rangle$
deconfined	perimeter	perimeter
Higgs	perimeter	**area**
confined	**area**	perimeter

Proposition 3.28 (screening)**.** *[heuristic] In the presence of matter in the fundamental representation, $\langle W_e(\gamma) \rangle$ obeys a perimeter law throughout the phase diagram, because a long string can lower its energy by breaking through the creation of a matter pair at cost independent of the string length.*

Interpretation 3.29 (the Fradkin–Shenker puzzle)**.** Proposition 3.28 removes the second column of the table's discriminating power: W_e cannot separate the Higgs row from the confined row. By the self-duality of Remark 3.1, the same argument with J in place of t — the term $-J \sum_\ell \sigma^x_\ell$ creates and moves m particles — screens the 't Hooft loop W_m as soon as $J \neq 0$. Away from the two axes neither loop separates the Higgs regime from the confined one, and nothing else does either: Fradkin and Shenker showed that the two are connected without a phase transition [FS79]. What remains sharp is the distinction between the deconfined phase and the rest, which is a statement about topological order (Proposition 3.24), not about a local or a loop order parameter.

Remark 3.30 (where this leads)**.** The three facts assembled here — no local order parameter (Theorem 3.8), the surviving observables are extended (Definition 3.15), and in the pure gauge theory the discriminating one is a loop (Interpretation 3.29) — are the reason the modern treatment of this phase diagram is phrased in terms of higher-form symmetry. The order parameter is promoted from a point to a loop,

$$\langle \phi(x) \rangle \longrightarrow \langle W(\gamma) \rangle,$$

and Landau's framework is extended rather than abandoned. Matter of unit charge breaks the one-form symmetry explicitly — this is the screening of Proposition 3.28 — which is why the Higgs and confined regimes are not separated. Developing that is beyond these notes.

3.9 Summary

1. A charged local field has vanishing expectation value in any gauge theory with compact gauge group, with $\langle \phi_v \rangle_h = \tanh h$ exactly, uniformly in volume, dimension and coupling (Theorem 3.8). The mechanism is that a local gauge transformation costs $O(h)$ rather than $O(hV)$.

2. Gauge fixing makes $\langle \phi \rangle \neq 0$ available but gauge dependent, hence not an observable (Example 3.12).

3. Charges are created by open strings; condensation is long-range order of the string, equivalently vanishing string tension (Definition 3.17).

4. Which condensates are possible is fixed algebraically: for $\mathbb{Z}_2$ exactly two, $\{1, e\}$ and $\{1, m\}$, i.e. Higgs and confinement (Proposition 3.22); both leave the same trivial theory, and with unit-charge matter the two regimes are connected without a transition.

5. The toric code's four-fold degeneracy is the count of $\mathbb{Z}_2$ holonomies around the two cycles of the torus — the global data that gauge fixing cannot remove (Proposition 3.24, Interpretation 3.25).

6. With fundamental matter the Wilson loop is screened and cannot separate Higgs from confinement; by self-duality neither can the 't Hooft loop once $J \neq 0$, and the two regimes are one phase. The sharp distinction is between the deconfined phase and the rest, i.e. topological order (Interpretation 3.29).

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